

STAC67: Regression Analysis

Lecture 2

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Covariance and Correlation

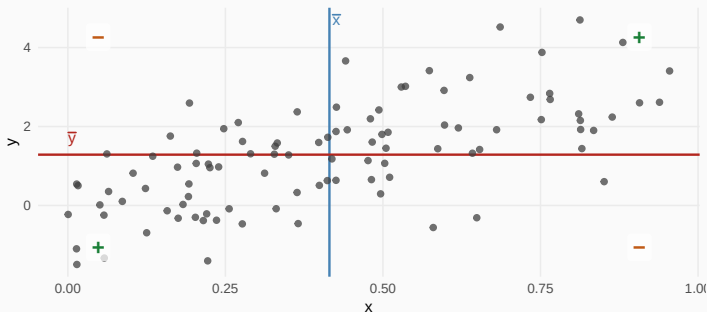
Covariance: $Cov(X, Y) = E((X - \mu_x)(Y - \mu_y))$, where $\mu_x = E(X)$, $\mu_y = E(Y)$

For $Z_x = \frac{X - \mu_x}{\sqrt{Var(X)}}$, $Z_y = \frac{Y - \mu_y}{\sqrt{Var(Y)}}$, the **correlation coefficient** is given by

$$Cov(Z_x, Z_y) = \rho_{xy} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

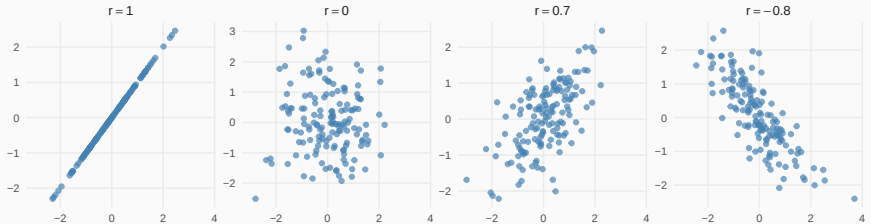
- $-1 \leq \rho_{xy} \leq 1$
- When the relationship is perfectly linear then $|\rho| = 1$.
- if two variables are independent then $\rho = 0$. (Note: the inverse does not hold)

Sample Covariance and Correlation



$$Cor(Y, X) = Cov(Z_y, Z_x) = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{y_i - \bar{y}}{s_y} \right) \left(\frac{x_i - \bar{x}}{s_x} \right)$$

Correlation



Question: what are main differences between correlation and regression model?

Test for Population correlation

When $\rho = 0$, and the joint distribution of (X, Y) is bivariate normal, and it can be shown that:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

has a Student's t distribution with $n - 2$ degrees of freedom

$$H_0 : \rho = 0 \quad vs \quad H_1 : \rho \neq 0$$

Testing procedure

- Calculating the observed value of t (call this t_{obs})
- Compute the p-value for the test

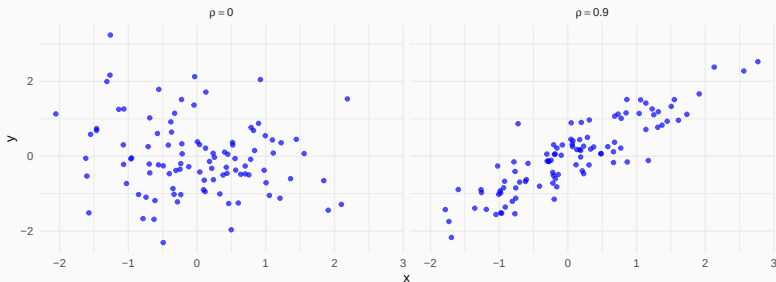
Simulation

```
library(mvtnorm)
set.seed(123)
sigma.1 <- matrix(c(1, 0, 0, 1), ncol = 2) # rho = 0
sigma.2 <- matrix(c(1, 0.9, 0.9, 1), ncol = 2) # rho = 0.9

X1 <- rmvnorm(100, mean = c(0, 0), sigma.1)
X2 <- rmvnorm(100, mean = c(0, 0), sigma.2)

library(ggplot2)
sim <- rbind(data.frame(lab = "rho == 0", x = X1[, 1], y = X1[, 2]),
             data.frame(lab = "rho == 0.9", x = X2[, 1], y = X2[, 2]))

ggplot(sim, aes(x, y)) +
  geom_point(size = 0.7, alpha = 0.7, colour = "blue") +
  facet_wrap(~ lab, labeller = label_parsed) +
  theme_minimal(base_size = 7)
```



Simulation

```
x <- X2[, 1]
y <- X2[, 2]
cor.test(x, y)

##
## Pearson's product-moment correlation
##
## data: x and y
## t = 18.766, df = 98, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.8327213 0.9209058
## sample estimates:
## cor
## 0.8844736

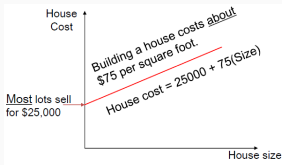
r <- cor(x, y)
t <- r * sqrt(length(x) - 2) / sqrt(1 - r^2)
t

## [1] 18.76559
```

Relationship between variables

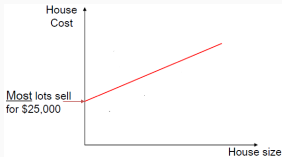
What factor or variable affects the price of house?

1) **Mathematical relation:** $Y = f(X)$, where X, Y are variables and f is a function



2) **Statistical relation:**

$$Y = f(X) + \epsilon$$



Data Collection for regression analysis

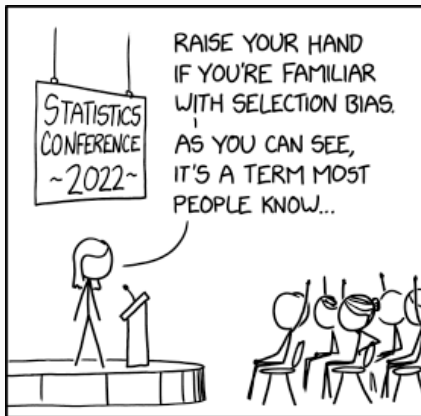
- **Observational study**

- Investigator has no control over the explanatory variables (X)
- Limitation: not adequate for cause-and-effect. A strong association does not necessarily mean a cause-and-effect relationship

- **Experiment**

- Investigator exercises control over the explanatory variables (X) through random assignment
- Random assignment balances out effect of other variables that might affect Y
- Gold standard for cause-and-effect conclusions

Who ends up in your data?



Getting into the sample is itself an outcome and rarely a random one...

xkcd.com/2618 (CC BY-NC 2.5)

The Regression Process

1. The researcher must clearly define the question(s) of interest in the study
2. The response variable Y must be decided on, based on the question of interest.
3. A set of potentially relevant covariates, which can be measured, needs to be defined.
4. Data is collected.
5. Model Specification.
6. Decide on a method for fitting the specified model
7. Fit the model - typically using software such as R
8. Examine the fitted model for violations of assumptions.
9. Conduct hypothesis testing for questions of interest.
10. Report the results from statistical inference.

Background Review: distributions

Simple Linear Regression

Suppose we have n observed pairs $(X_i, Y_i), i = 1, \dots, n$.

Assumptions

1. A linear relationship

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

- Y_i is the value of the **response variable** in the i th trial.
- X_i is a known constant, namely, the value of the **predictor variable** in the i th trial.
- β_0, β_1 : regression model coefficients

β_0 :

β_1 :

- 2. ϵ_i are random errors with zero mean, $E(\epsilon_i) = 0$, with common variance, $Var(\epsilon_i) = \sigma^2$, and pairwise independent.

Important Features

In the simple linear model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

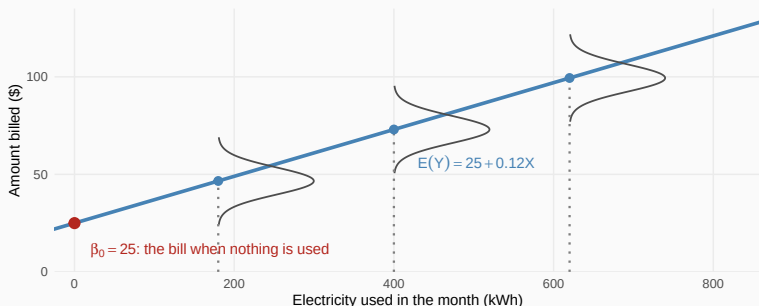
1. The response variable Y is a sum of two terms:
2. $E(Y_i) = \beta_0 + \beta_1 X_i$, where $E(Y_i)$ is a shortcut for $E(Y_i|X_i)$, the mean of Y when $X = X_i$
3. $Var(Y_i) = \sigma^2$, where $Var(Y_i)$ is a shortcut for $Var(Y_i|X_i)$, the variance of Y when $X = X_i$
4. The outcomes Y_i are pairwise independent because ϵ_i are pairwise independent.

Important Features

Example

Y = the monthly electricity bill, X = electricity used that month (kWh)

- Regression function: $E(Y) = 25 + 0.12X$
- $\beta_1 = 0.12$ indicates:
- $\beta_0 = 25$ indicates:



Exercise

Exercise

- The regression model applies with $\beta_0 = 100$, $\beta_1 = 20$, and $\sigma^2 = 25$. An observation on Y will be made for $X = 5$.
- a. Can you state the exact probability that Y will fall between 195 and 205? Explain.
- b. If the normal error regression model is applicable, can you now state the exact probability that Y will fall between 195 and 205? If so, state it.